

Nhat Hung Nguyen

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EDUCATION

Bachelor of Applied Science and Engineering (B.A.Sc) in Electrical Engineering

Toronto Metropolitan University (Formerly known as Ryerson University)

Sep. 2021 - May 2026

- Relevant courses: Electronic Circuits I/II, Software Systems, Electric and Magnetic Fields, Microprocessor Systems, Signal and System, Energy Conservation, Power Electronics, Power System Analysis, Advanced Electric Drives, Low Power Digital Integrated Circuit

SKILLS

Programming/Simulation || AutoCAD, Python, Java, Excel, MATLAB, C, Multisim, Arduino, VS Code, Revit, ETAP

Tools || Oscilloscope, Function Generator, Power Supply, Signal Generator, Control System, Digital Multimeter, Insulation Resistance Tester, DC Power Supplies, Wire Strippers and Cutters, ESP32, ESP8266, OP-AMPS, FPGA

Other || Microsoft Office [Teams, Word, Publisher, Powerpoint, Excel], Analytical, Organization, Communication, Teamwork, Multitasking, Enthusiastic, Problem-solving, Leadership, Time Management, Passion for Learning

WORK EXPERIENCE

Maintenance Worker | CRM Rubber Manufacturers - Brantford, ON

May 2026 - Present

- Conducted routine hourly inspections of production machinery to verify normal operating conditions.
- Prepared operational reports and communicated equipment irregularities to management to support preventive maintenance activities.
- Supported troubleshooting efforts by observing system performance and reporting irregularities.

Cashier | Coco Fresh Tea & Juice - Toronto, ON

June 2024 - April 2026

- Demonstrated excellent time management by efficiently handling high volumes of customer transactions during peak hours, while maintaining accuracy and a positive attitude.
- Collaborated effectively within a high-volume team environment to maintain efficient customer service during peak operating periods.

RELEVANT PROJECT

Smart Bartender Ordering System | *Python, FastAPI, ESP8266, Arduino, REST APIs, HTML/CSS/JavaScript, Wi-Fi*

- Designed and implemented a full-stack automated beverage dispensing system integrating a FastAPI (Python) backend, ESP8266 microcontroller, and web-based user interface.
- Developed RESTful APIs and JSON-based Wi-Fi communication to manage drink orders, system state, and synchronized software-to-hardware execution.
- Programmed ESP8266 firmware (Arduino C/C++) to parse backend instructions and translate drink recipes into timed relay control signals for pump-based liquid dispensing.
- Implemented a machine learning recommendation model that analyzes historical user preferences to suggest personalized beverages.

Multi-Stage Analog Circuit Design and Waveform Generator | *BJT, OP-AMPS, MOSFETS, Function Generator*

- Developed, simulated, analyzed, implemented, and tested a transistor amplifier with multiple stages operating on a single supply, ensuring it met a predefined set of specifications.
- Conceived, simulated, executed, and validated a Linear Voltage-Controlled Multi-function Waveform Generator (VCFG) circuit - primarily focused on delivering a practical demonstration of knowledge through waveform generation techniques.

Embedded Control Systems | *C++, Microcontroller, Arduino, ESP32, DMM, Sensors*

- Developed multiple embedded system prototypes integrating RGB LEDs, switches, servo motors, ultrasonic sensors, LCD displays, and 7-segment displays.

VHDL/Verilog-Based Arithmetic Processor Design | *Verilog, Quartus, VHDL, RTL Design*

- Designed and implemented a custom arithmetic processor featuring an ALU and control unit using VHDL and Verilog.